

Noise-Factorized Disentangled Representation Learning for Generalizable Motor Imagery EEG Classification

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Abstract—Motor Imagery (MI) Electroencephalography (EEG) is one of the most common Brain-Computer Interface (BCI) paradigms that has been widely used in neural rehabilitation and gaming. Although considerable research efforts have been dedicated to developing MI EEG classification algorithms, they are mostly limited in handling scenarios where the training and testing data are not from the same subject or session. Such poor generalization capability significantly limits the realization of BCI in real-world applications. In this paper, we proposed a novel framework to disentangle the representation of raw EEG data into three components, subject/session-specific, MI-task-specific, and random noises, so that the subject/session-specific feature extends the generalization capability of the system. This is realized by a joint discriminative and generative framework, supported by a series of fundamental training losses and training strategies. We evaluated our framework on three public MI EEG datasets, and detailed experimental results show that our method can achieve superior performance by a large margin compared to current state-of-the-art benchmark algorithms.

Index Terms—Disentangled representation learning, motor imagery EEG, generalization, brain-computer interfaces.

I. INTRODUCTION

RECENT advances in Brain-Computer Interfaces (BCIs) have enabled a wide range of applications, in the field of affective computing [1], gaming [2], and rehabilitation engineering [3]. In particular, for rehabilitation purposes, emerging BCI methods have enabled intuitive controls over prosthetics for amputees, as well as personalized therapeutic training for patients after stroke [4].

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Electroencephalography (EEG) is a non-invasive brain functional imaging modality that has been widely used in BCI research. Compared to other non-invasive modalities, such as fMRI and fNIRS, EEG can provide much higher temporal resolution, with good portability and low acquisition cost. In the context of BCI applications, EEG has been instrumental not only in rehabilitation but also in the development of assistive technology, enhancing accessibility and user interaction. Motor Imagery (MI) represents one of the prevalent BCI paradigms, involving the mental rehearsal of specific body part movements. This cognitive process triggers a sequence of EEG rhythm changes in the sensorimotor areas, characterized by event-related desynchronizations (ERD) at the onset and followed by event-related synchronizations (ERS) upon completion of the imagined movement. These rhythm change patterns, unique to the imagined body part, offer an intuitive means for controlling assistive tools and devices, providing a robust foundation for various BCI applications.

Over the past two decades, considerable research efforts have been dedicated to extracting discriminative features to distinguish different MI tasks [5], [6], [7], [8], [9]. Conventional feature engineering methods are mostly aimed at extracting discriminative features that are relevant to ERD/ERS patterns based on human expertise and domain knowledge [5], [6], [7]. In particular, the filter-bank common spatial pattern algorithm, FBCSP [6], which maximizes inter-class discrepancies through multiple-frequency-band-based spatial filtering, has been one of the benchmark algorithms for MI EEG classification.

With the rapid development of deep learning algorithms, the application of advanced computational models on MI EEG classification has seen a surge of research activity [8], [9], [10], [11], [12], [13], [14], [15]. The core design of these computational models is driven by the need for extracting discriminative temporal, spatial, and frequency features, or combinations. They have demonstrated good MI classification performance surpassing conventional methods. However, these models intrinsically suffer from the overfitting issue, as the underlying distributions of training and testing data are usually not the same, and the training set is usually poor in diversity and contaminated by noises. In fact, the characteristics of raw EEG data pertain to a variety of factors; especially, they change significantly across different subjects and sessions. Specifically, the inter-subject variability is often more pronounced than the inter-session variability [16]. The vanilla version of these models cannot overcome the discrepancies, thus leading to failure in cross-subject or cross-session validation settings [17].

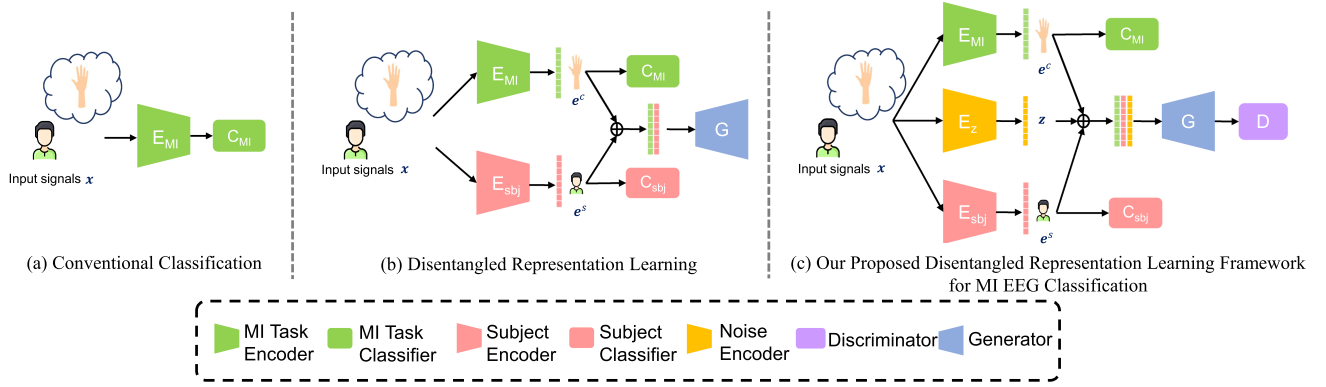


Fig. 1. Comparison between the conventional classification training and our proposed disentangled representation learning framework. (a) For conventional training, the input signals are fed through a network for classification in an end-to-end manner, with the backbone E_{MI} extracting MI-task related features, and C_{MI} performing classification on top of the features. (b) For conventional disentangled representation learning, a dual-encoder autoencoder structure is applied to disentangle subject and motor-imagery related information in each encoder. (c) For our proposed disentangled representation learning framework, the input signal x is projected to the latent feature space with three components, MI-task-specific e^c , subject-specific e^s , and noise vector z , by E_{MI} , E_{sbj} , and E_z respectively. C_{MI} and C_{sbj} are applied for classification training of e^c and e^s , to ensure the discriminativeness. Subsequently, generator G generates synthetic samples with the input of latent vectors. Meanwhile, another discriminator D enables adversarial training to ensure the realism of generated samples.

Hitherto, the generalization issue of MI EEG classification has been raised in the literature [14], [18], [19], [20], [21]. However, most current solutions (e.g., unsupervised domain adaptation, UDA) [18], [19] assume the availability of target data in the training phase, which is not an ideal countermeasure for enabling real-world applications. In fact, ideally, a robust model is expected to perform well on any data from novel subjects or sessions without any further training/fine-tuning.

Recent research on domain generalization (DG) has typically probed investigations into the aforementioned problem with the goal of generalizing to unseen domains, although mainly focused in the computer vision field [22]. Our work was inspired by one recent line of DG works that is based on disentangled representation learning [23]. This line of work disentangles domain-specific features apart from semantic-specific features, so that the classification is robust against domain shifts. One popular solution for realizing disentanglement is to apply a dual-encoder single-decoder structure. The dual-encoder disentangles two groups of features by discriminative training (mostly by classification), whereas the single-decoder ensures the completeness of the learned feature by generative training (mostly by reconstruction).

a) Unlike image reconstruction tasks, where basically the reconstruction is mostly based on per-pixel reconstruction errors [24], [25], [26], applying per-timestamp reconstruction of the EEG times series is not feasible. In most cases, EEG presents chaotic dynamics and a low signal-to-noise ratio in the temporal domain, so achieving the similarity in the temporal domain may just resemble the noises.

b) EEG is complex, the characteristics of which are not only manifested by the imagination task, and subject features but also by a variety of other factors, such as artifact contamination, temporal variances, etc. These factors can hardly be eliminated by preprocessing. Hence, only applying a dual-encoder to disentangle subject/session and task information is not enough.

c) Although there have been quite a few works/frameworks proposed for the disentanglement of images, applying them to EEG is not trivial, especially with the challenges mentioned above. It remains open to developing an effective framework

together with successful training strategies to disentangle subject-specific and MI-specific features.

Therefore we propose a novel framework to perform representation disentanglement of raw MI EEG data, thus facilitating a generalized classification model in cross-subject/session settings. As shown in Fig. 1, the latent features extracted from raw EEG signals are disentangled into three components, subject-specific features, MI-task-specific features, and noises. To achieve the disentanglement, we provide deep insights into the architecture design, loss functions, as well as training strategies for EEG-based generative models. In line with the key challenges, our contributions are as below,

a) *Noise-Factorized Disentanglement Architecture*: Different from the existing commonly applied dual-encoder structure, another branch encoding the noise-related information is incorporated. Our experiment results have validated the effectiveness of such architectural design, which shows the proposed method surpasses the performance of the dual-encoder version.

b) *Novel Training Losses and Strategies*: In accordance with triple-encoder architecture, we developed a series of training losses and strategies to effectively disentangle subject-related, MI-task-related, and noise-related information. Meanwhile, we provided detailed experiments on validating the realism of synthetic EEG samples generated by the decoder.

c) *Cross-Subject/Session Generalization*: Extensive experiments were conducted to validate the representativeness of the feature encoded by each branch, as well as the realism of generated synthetic samples. Extensive experiment results have shown that our method achieved significantly better classification than current state-of-the-arts among three public MI EEG datasets.

II. RELATED WORKS

A. Deep Learning Based EEG Brain-Computer Interfaces

Deep learning (DL) models have become popular in the field of BCI. Existing MI EEG studies employed deep learning methods on a variety of architectures, such as convolutional

neural network [8], [9], recurrent neural network [10], [11], [12], [27], graph neural network (GNN) [28], [29]. These DL models scale well to large datasets and can simultaneously learn intricate high-dimensional features from raw signals in an end-to-end manner [8].

However, on the other hand, the shortcomings of DL models become apparent when the training data size is small. Without a massive amount of training data, the testing data sampled from novel subjects or sessions might exhibit a distribution diversified from the training data. In fact, the size of current MI EEG datasets is limited [30], caused by several issues such as cumbersome calibration procedures, uncertainty in annotations due to variability in participants' responses to MI tasks [31], and data privacy issues [32]. These limitations unfortunately lead to the overfitting of DL models.

Recent research has attempted to apply domain adaptation techniques to address this issue [18], [33]. However, most current domain adaptation paradigms assume the availability of testing data distribution during model training, which is not ideal for the practical deployment of BCI algorithms. It remains an open problem to generalize models to unseen subjects or sessions for MI EEG classification.

B. Generalization Solutions for EEG Classification

Existing works on the generalization capability of EEG classification mainly focus on sufficient data augmentation, and effective representation learning. Conventional methods for EEG augmentations, e.g., noise addition, frequency shifts [13], on temporal or frequency domain. More advanced methods apply deep generative models like variational autoencoder (VAE) or generative adversarial network (GAN) to synthesize novel data based on sampling from learned data distributions [34], [35], [36]. These existing methods demonstrated promising results in dealing with EEG classification problems. However, there remains no guarantee that the generated data would be of high quality and well representative for each class. In fact, the generated data are usually used for training simultaneously with real data or finetuning. Furthermore, since the distribution is learned from data from existing subjects/sessions, the cross-subject/session performance by vanilla deep generative models is still limited.

On the other hand, representation learning mainly focuses on abstracting subject/session invariant features from raw EEG data [37], [38], [39]. For instance, Ma et al. [37] proposed a structure that is composed of a set of subject-shared unbiased weights and subject-specific biased weights through adversarial training, where the unbiased weights are expected to derive subject-invariant features. Autthasan et al. [39] proposed a multi-task learning framework, MIN2NET, for cross-subject EEG generalization. The framework integrates deep metric learning into a multi-task autoencoder to learn a compact and discriminative latent representation from EEG and perform classification simultaneously.

More generally, the goal of ensuring cross-subject/session generalization performance is in line with the motivations of recent DG works [22] in the computer vision field that aims to generalize to unseen domains. In particular, one representative line of DG solutions is to derive domain-invariant features through disentangled representation learning, which is discussed below.

C. Disentangled Representation Learning

Disentangled representation learning aims to disentangle the features into domain-shared and domain-specific parts for inter-domain generalization purposes [22], [23], [26], benefit from simultaneously generative and discriminative modeling, with the generative module learning compact and complete features, whereas the discriminative one ensures the separability of learned features. For instance, Tran et al. [23] proposed a pose-invariant face recognition framework that uses hard-coded pose features and network-learned identity features to synthesize facial images from different viewpoints. Zheng et al. [26] proposed a pose-invariant person re-identification framework, with appearance (id-related) and structure (id-irrelevant) features learned simultaneously. The success of this framework is mainly attributed to its training strategies including self/cross-identity generation and discriminative identity feature learning.

Drawing parallels to human-centric data, recent studies have sought inspiration from disentangled representation learning to extract subject-invariant features from physiological or behavioural signals. For instance, Aberman et al. [40] introduced a cross-subject reconstruction strategy for human motion retargeting. Similarly, works by Gu et al. [41] and Han et al. [42] presented solutions to derive subject-invariant features from physiological signals using cascaded cross-modal transfer and domain adversarial learning, respectively. Specifically, within the MI-EEG domain, Autthasan et al. [39] introduced MIN2NET, a multi-task learning framework for cross-subject EEG generalization. This framework combines deep metric learning with a multi-task autoencoder to simultaneously derive a compact, discriminative latent representation from EEG and perform classification.

However, these disentangled frameworks cannot be adopted to MI EEG classification directly. The EEG time series are fundamentally different from 2D images, where EEG is expanded across the time axis with low signal-to-noise-ratio, and is also semantically meaningful in the frequency domain. Furthermore, the inter-subject/session variations in EEG are non-intuitive compared to poses [23] or styles [24] in computer vision, thus representing them as categorical values or one-hot vectors might not be as effective [23]. In addition, most of the generative methods learn image generation either from random noise or feature vectors only, which makes it difficult to control the output and introduce data variations.

In contrast, our framework learns the MI-specific features, the subject-specific features and the other factors (denoted as noise) directly from EEG signals in an end-to-end manner. The framework is capable of generating realistic and diverse cross-subject motor imagery EEG signals while maintaining high classification accuracy on the disentangled features.

III. METHODS

A. Problem Definition and Preliminaries

Without loss of generality and for simplicity, we only consider cross-subject settings in the methodology section, where each subject is considered as an individual domain. The cross-session settings can be easily sorted out by viewing each session as an individual domain.

Given the i th sample, we denote it as (x_i, y_i, s_i) , where the input data is denoted as $x_i \in \mathbb{R}^{ch \times t}$ (ch refers to the channel

number, t the sampling number of each segment), the ground truth MI class as $y_i \in \mathbb{R}$, and the subject index as s_i . The sets of training and testing subjects are denoted as $\mathcal{S}_{tr}$ and $\mathcal{S}_{te}$, respectively. Subsequently, the training dataset can be represented as $\{\mathbf{x}_i, y_i, s_i | s_i \in \mathcal{S}_{tr}\}$ and the testing $\{\mathbf{x}_i, y_i, s_i | s_i \in \mathcal{S}_{te}\}$. In cross-subject settings, we set $\mathcal{S}_{tr} \cap \mathcal{S}_{te} = \Phi$, which means the data from testing subjects are totally unseen during training.

As discussed above, the characteristics of data $\mathbf{x}$ are affected by not only the MI task and the subject individual inner-nature, but also other random factors. In this regard, we would like to derive a model that can effectively learn latent representations specific to these three groups, so that the MI-specific representations can facilitate the generalization in cross-subject settings.

In practice, we introduce a series of computational models E_{MI} , E_{sbj} , and E_z that are expected to extract MI-specific feature e^c , subject-specific feature e^s , and noise z respectively. To ensure the compactness and completeness of the derived features, a generative function G is introduced to convert the features back to the original signal space, as shown in Fig. 1. To ensure disentangled representation learning under the above preliminaries, the requirements for effective model training are listed as follows,

- **Realism:** The generator is expected to generate samples that can resemble the original raw EEG signals. In this way, the extracted features e^c , e^s , and z are expected to fully represent the distribution of EEG data sets. This ensures the compactness of these features, without collapsing to any trivial solutions.
- **Diversity:** The generative modeling should also uncover those “unseen” data by promoting the diversity of the generated data. Especially, the noise vector z , which encodes the variations of EEG signals other than MI tasks and subjects, is expected to take effect.
- **Discriminateness:** More importantly, the extracted features e^c , e^s , and z should capture their branch-unique information, encoding MI-task-specific, subject-specific, as well as other random factors in EEG representations.

B. Framework Overview

The overall framework of our proposed method is shown in Fig. 1. On top of the encoders E_{MI} , E_{sbj} , E_z and the generator G , we also apply two classifiers C_{MI} , C_{sbj} , and a discriminator D . The E_{MI} , E_{sbj} , E_z encode task-specific, subject-specific, and noises respectively. Subsequently, C_{MI} , C_{sbj} are aimed at classifying the MI classes and subject identities based on the features extracted from E_{MI} and E_{sbj} , respectively. The generator G converts the latent features e^c , e^s , and z back to the original signal space by synthesizing fake samples $\hat{\mathbf{x}}$, i.e. $\hat{\mathbf{x}} = G(e^c, e^s, z)$. The D is used to distinguish real and synthetic samples with adversarial training. Below, we introduce the fundamental training losses and training strategies to enable an effective disentanglement of e^c , e^s , z .

C. Self-Reconstruction and Self-Generation

1) **Self-Reconstruction by Conditional Variational Autoencoder:** To enable the generative modeling of the EEG samples, we adopted the structure of conditional variational autoencoder (C-VAE) [43]. The pipeline of C-VAE training is shown in Fig. 2. The optimization for the distribution modeling of C-VAE is

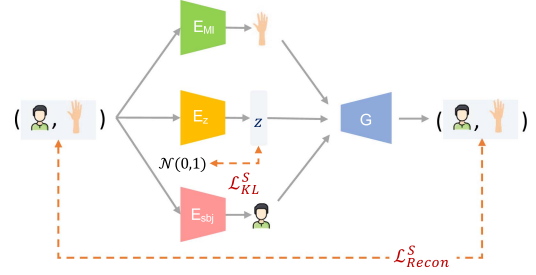


Fig. 2. Illustrations of self-reconstruction enabled by conditional variational autoencoder.

realized by the following evidence lower bound [43],

$$\min_{E, G} KL(q(z|\mathbf{x}, y, s) || p(z|y, s)) - \mathbb{E}_{q(z|\mathbf{x}, y, s)} [\log p(\mathbf{x}|z, y, s)], \quad (1)$$

where z refers to the latent distribution conditioned on $\mathbf{x}$, y , and s . p represents the realistic distribution, whereas $q(z|\mathbf{x}, y, s)$ represents the posterior distribution our model learns and $p(z|y, s)$ is the prior distribution. In practice, q is modeled by E_z . The term $KL(q(z|\mathbf{x}, y, s) || p(z|y, s))$ aims to align p and q by the Kullback–Leibler (KL) divergence, whereas $\mathbb{E}_{q(z|\mathbf{x}, y, s)} [\log p(\mathbf{x}|z, y, s)]$ reflects the reconstruction of the original signal by the latent z .

Reconstruction Loss: To maximize the likelihood of $\mathbb{E}_{q(z|\mathbf{x}, y, s)} [\log p(\mathbf{x}|z, y, s)]$ in (1), the generated samples $\hat{\mathbf{x}}$ needs to resemble the input $\mathbf{x}$, which can be enforced by applying reconstruction losses. Original image reconstruction works tend to apply $\mathcal{L}_2$ or $\mathcal{L}_1$ loss to estimate the reconstruction error across pixels. Differently, for EEG time series, our preliminary experiments demonstrated that applying $\mathcal{L}_2$ or $\mathcal{L}_1$ in the temporal domain would ignore some fundamental characteristics of EEG signal in the frequency domain (see discussions in Section IV-F4). Based on this observation, we instead calculate the reconstruction loss on the time-frequency domain after short-term Fourier transformation (STFT) [44]. The reconstruction loss between $\mathbf{x}_i$ and $\hat{\mathbf{x}}_i = G(e_i^c, e_i^s, z_i)$ are denoted as below,

$$\begin{aligned} \mathcal{L}_{Recon}^S &= \mathbb{E}_i [\mathcal{L}_{stft}(\mathbf{x}_i, \hat{\mathbf{x}}_i)] \\ &= \mathbb{E}_i \left[\frac{1}{M} \sum_{m=1}^M [\mathcal{L}_{sc}^{(m)}(\mathbf{x}_i, \hat{\mathbf{x}}_i) + \mathcal{L}_{mag}^{(m)}(\mathbf{x}_i, \hat{\mathbf{x}}_i)] \right], \end{aligned} \quad (2)$$

where M represents the number of different resolutions. We used $M = 3$ in our experiments. $\mathcal{L}_{sc}$, $\mathcal{L}_{mag}$ denote the spectral convergence loss and the log magnitude loss respectively, which are formulated as below,

$$\begin{aligned} \mathcal{L}_{sc}^{(m)}(\mathbf{x}_i, \hat{\mathbf{x}}_i) &= \frac{\| |\text{STFT}(\mathbf{x}_i)^{(m)}| - |\text{STFT}(\hat{\mathbf{x}}_i)^{(m)}| \|_F}{\| |\text{STFT}(\mathbf{x}_i)^{(m)}| \|_F}, \\ \mathcal{L}_{mag}^{(m)}(\mathbf{x}_i, \hat{\mathbf{x}}_i) &= \frac{\| \log |\text{STFT}(\mathbf{x}_i)^{(m)}| - \log |\text{STFT}(\hat{\mathbf{x}}_i)^{(m)}| \|_1}{K}, \end{aligned} \quad (3)$$

where $\| \cdot \|_F$ and $\| \cdot \|_1$ represent the Frobenius and $\mathcal{L}_1$ norms. $|\text{STFT}(\cdot)|$ represents the magnitude under resolution M and K denotes the number of elements in current magnitude.

Noise Distribution Modeling Loss: The term $KL(q(z|x, y, s)||p(z|y, s))$ in (1) is aimed for explicitly modeling the distribution of z to $\mathcal{N}(0, 1)$ by minimizing the KL divergence between both. Under such, the learned latent z can represent the factors independent of subject and MI tasks.

The basic loss functions are denoted as below,

$$\mathcal{L}_{KL}^S = \mathbb{E}_i[\mathcal{L}_{KL}^z(x_i)] = \mathbb{E}_i[KL(E_z(x_i) || \mathcal{N}(0, 1))], \quad (4)$$

where KL represents the function that measures the KL distribution divergence. In practice, we applied the reparameterization trick to learn the distribution in a deterministic end-to-end manner. This trick allows us to sample from a distribution by sampling from a standard noise distribution and transforming it using learned parameters, thereby making the stochastic computation tractable for gradient-based optimization [43].

2) Self-Generation by Generative Adversarial Training: To further increase the realism of the generated signal, we applied generative adversarial training, as in (5). We chose to use the WGAN-GP [45], which applies the Earth-Mover distance (EMD) to measure the similarity between real and synthetic samples. This has been proven to be capable of solving unstable training issues such as vanishing gradient, gradient instability, and imbalance of diversity in conventional GANs. In addition, WGAN-GP uses an effective gradient penalty term instead of weight clipping to enforce the Lipschitz constraint [45]. Its update goal is as shown below,

$$\min_G \max_D \mathbb{E}[D(x)] - \mathbb{E}[D(\hat{x})] - \lambda \mathbb{E}[\|\nabla_{\tilde{x}} D(\tilde{x})\|_2 - 1]^2, \quad (5)$$

where $\tilde{x}$ is a mixup of real sample x and synthetic sample $\hat{x}$ by parameter α , i.e., $\tilde{x} = \alpha x + (1 - \alpha)\hat{x}$. λ represents the weight of the gradient penalty term [45].

Adversarial Losses: In line with (5), the adversarial losses are applied with a discriminator D to distinguish real and synthetic images. The losses are formulated as below,

$$\begin{aligned} \mathcal{L}_D(x, \hat{x}) &= \mathbb{E}_{x_i \sim p(x)} D(\hat{x}_j) - D(x_i) \\ &\quad + \lambda (\|\nabla D(\alpha x_i + (1 - \alpha)\hat{x}_j)\|_2 - 1)^2 \\ &\quad + \epsilon (D(x_i) + D(\hat{x}_j))^2, \end{aligned} \quad (6)$$

$$\mathcal{L}_G(\hat{x}) = \mathbb{E}_{x_i \sim p(x)} - D(\hat{x}_j) + \epsilon (D(x_i) + D(\hat{x}_j))^2, \quad (7)$$

where x and $\hat{x}$ represent real and synthetic samples, respectively. λ is the weight for the gradient penalty term. α is a vector with the same shape as the data filled with random numbers from a uniform distribution between 0 and 1. The ϵ term is the weight of a drift penalty term to keep the discriminator centered at 0 [46]. Empirically, λ was set as 10 and ϵ as 0.001.

Overall, regarding self-generation (Fig. 3), a random distribution n_i from $\mathcal{N}(0, 1)$ and the resemblance between the real samples and the generated synthetic samples $G(e_i^c, e_i^s, n_i)$ is ensured by the following losses by alternative optimization,

$$\begin{aligned} \mathcal{L}_{Gen}^S &= \mathbb{E}_i[\mathcal{L}_G(G(e_i^c, e_i^s, n_i))], \\ \mathcal{L}_{Dis}^S &= \mathbb{E}_i[\mathcal{L}_D(x_i, G(e_i^c, e_i^s, n_i))]. \end{aligned} \quad (8)$$

Latent-Consistency Loss: Furthermore, the latent consistency loss is leveraged in the self-generation procedure to promote the diversity of sample characteristics enabled by the changes of latent vectors. In the literature, a wide range of generative networks

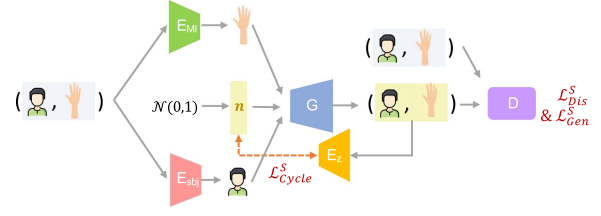


Fig. 3. Illustrations of self-generation enabled by generative adversarial training.

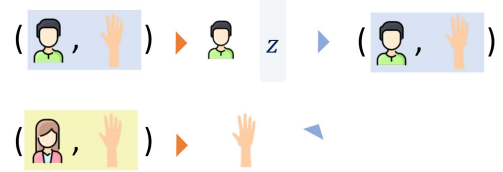


Fig. 4. Illustrations of Cross-Subject Intra-Class Generation. For each sample, another one from the same MI class yet different subjects is selected to provide the MI feature e^c . With the subject-specific feature e^s and noise vector z extracted from the original sample and z from the other sample, we aim to reconstruct the original sample, thus achieving the similarity of e^c across different subjects but the same MI task.

such as conditional variational autoencoder GAN (cVAE-GAN) and conditional GANs (cGAN) have been proposed to generate synthetic data under a condition. However, these models tend to maximize the generation quality at the expense of data variations, as the generator learns to largely ignore the random noise vector when conditioned on any relevant context [24], [25], [47], [48]. Furthermore, multiple studies also reported that the models had shown more stable training when the input noise vectors were ignored [25], [47], [48]. We, therefore, utilize the idea of BicycleGAN [25] to enforce our model to produce both diverse and realistic MI EEG data. The loss is formulated as below,

$$\mathcal{L}_{Cycle}^S = \mathbb{E}_i[\|n_i - E_z(G(e_i^c, e_i^s, n_i))\|_1]. \quad (9)$$

3) Discriminative Training: Meanwhile, to ensure the discriminativeness of e^c and e^s with respect to MI task y and subject identity s , The Cross Entropy losses, $\mathcal{L}_{mi}$ and $\mathcal{L}_{sbj}$ are enforced as below,

$$\begin{aligned} \mathcal{L}_{MI} &= \mathbb{E}_i[\mathcal{L}_{mi}(x_i, y_i)], \\ \mathcal{L}_{Sbj} &= \mathbb{E}_i[\mathcal{L}_{sbj}(x_i, s_i)]. \end{aligned} \quad (10)$$

D. Cross-Generation

Only applying classification loss (Sec III-C3) is not enough for simultaneous generative and discriminative training [26]. To further enhance the discriminativeness of the e^c and e^s , we applied the cross-generation training strategies, as inspired by several existing works [26].

1) Cross-Subject Intra-Class Generation: To ensure generation similarity, we first perform a cross-subject intra-class generation task between pairs of EEG data with the same MI label but different subject labels. The pipeline is shown in Fig. 4. The generator learns to reconstruct x_i with the MI feature vector of a different data x_j under $y_i = y_j, s_i \neq s_j$. It is achieved by optimizing the STFT loss $\mathcal{L}_{stft}(x_i, G(e_j^c, e_i^s, z_i))$. This same-MI yet cross-subject generation task encourages the MI encoder

Algorithm 1: Proposed Training Framework for Disentangled Representation Learning of MI EEG.

Input: Training set $\{x_i, y_i, s_i | s_i \in \mathcal{S}_{tr}\}$; Initialized models $E_{MI}, E_{sbj}, E_z, G, D, C_{MI}, C_{sbj}, T_{MI}, T_{sbj}$; Initialized r_D, r_G as $+\infty$; Hyper-Parameters $\eta, \{w_k\}_{k=1}^{12}$.

Output: Optimized models E_{MI} and C_{MI} .

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1: for t < Pretrain Iterations    ▷ Teacher Model Training
2:   Sample a batch from the real data,  $\{x_i, y_i, s_i\}_{i=1}^B$ .
3:   Calculate  $\mathcal{L}_{MI}, \mathcal{L}_{sbj}$  based on  $T_{MI}, T_{sbj}$ .
4:   Update  $T_{MI}, T_{sbj}$  based on  $\mathcal{L}_{MI}, \mathcal{L}_{sbj}$ .
5: end for
6:
7: for t < Max Iterations        ▷ Training Start
8:   Sample a batch from the real data,  $\{x_i, y_i, s_i\}_{i=1}^B$ .
9:   Sample a batch of noise vector  $\{n_i\}_{i=1}^B \sim \mathcal{N}(0, 1)$ .
10:  if  $r_D > \eta \cdot r_G$           ▷ Update D
11:    Do Self-Generation (Section III-C2), Calculate  $\mathcal{L}_{Dis}^S$ .
12:    Do Cross-Generation (Section III-D), Calculate  $\mathcal{L}_{Dis}^C$ .
13:    Calculate  $\mathcal{L}_{Dis}$ , Update D based on  $\mathcal{L}_{Dis}$ .
14:  else                        ▷ Update  $E_{MI}, E_{sbj}, C_{MI}, C_{sbj}, G$ 
15:    Do Self-Reconstruction and Self-Generation (Section III-C)
16:    Do Cross-Generation (Section III-D)
17:    Calculate  $\mathcal{L}_{Gen}$ , Update  $E_{MI}, E_{sbj}, C_{MI}, C_{sbj}, G$  based on  $\mathcal{L}_{Gen}$ .
18:  end if
19:  Update  $r_G, r_D$ .
20: end for

```

to pull the latent representation of the same classes closer to reduce the intra-class feature variations. The loss functions are as below,

$$\mathcal{L}_{Gen}^{CI} = \mathbb{E}_{y_i=y_j, s_i \neq s_j} [\mathcal{L}_{stft}(x_i, G(e_i^c, e_i^s, z_i))]. \quad (11)$$

2) Cross-Subject Cross-Class Generation: To make sure that real and synthetic samples resemble each other when e^c and z are switched across subjects/tasks, adversarial losses are enforced in the subsequent cross-subject cross-class generation tasks. We, therefore, apply cross-subject cross-class generation as shown in Fig. 5.

First of all, given two EEG signals x_i and x_j with different MI labels and different subject labels $y_i \neq y_j, s_i \neq s_j$, the generator tries to generate new realistic data with $G(e_i^c, e_j^s, z_k) \rightarrow \hat{x}_i^c$, where k is a random index within the current batch.

Adversarial Losses: We continue to apply the generative adversarial loss to ensure realism.

$$\begin{aligned} \mathcal{L}_{Gen}^{CC} &= \mathbb{E}_{y_i \neq y_j, s_i \neq s_j} [\mathcal{L}_G(G(e_i^c, e_j^s, z_k))], \\ \mathcal{L}_{Dis}^{CC} &= \mathbb{E}_{y_i \neq y_j, s_i \neq s_j} [\mathcal{L}_D(x_i, G(e_i^c, e_j^s, z_k))]. \end{aligned} \quad (12)$$

Cycle-Consistency Losses: Furthermore, the generated EEG signal $\hat{x}_i^c$ needs to be enforced to retain the information of the MI feature vector e_i^c from x_i and subject feature vector s_t from x_t . Therefore, additional $\mathcal{L}_{Cycle}^C$ is enforced to achieve the

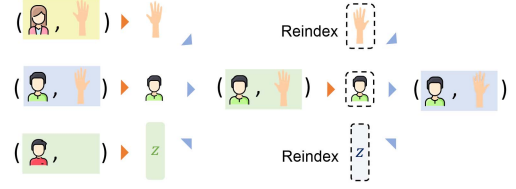


Fig. 5. Illustrations of Cross-Subject Cross-Class Generation. For each sample, the subject-specific e^s is extracted. Meanwhile, other samples from different subjects and different classes are randomly selected to provide the MI feature e^c , whereas a randomly selected sample provides the noise vector z . The synthetic sample is generated from these three latent vectors. Afterward, we continue to extract z and e^s from all the synthetic samples to reconstruct the original sample, thus enforcing cycle consistency.

consistency of latent features.

$$\begin{aligned} \mathcal{L}_{CycleE}^{CC} &= \mathbb{E}_{y_i \neq y_j, s_i \neq s_j} [\| (e_i^c, E_{MI}(e_i^c, e_j^s, z_k)) \|_1 \\ &\quad + \| e_j^s, E_{sbj}(e_i^c, e_j^s, z_k) \|_1]. \end{aligned} \quad (13)$$

Similarly, we calculate $\hat{e}^c, \hat{e}^s, \hat{z}$ from the cross-generated samples $\hat{x}^c$. Based on these, we re-index these vectors based on their original sample order, as shown in Fig. 5. Subsequently, we apply the $\mathcal{L}_{stft}$ loss to ensure cycle consistency.

$$\mathcal{L}_{CycleX}^{CC} = \mathbb{E}_i [\mathcal{L}_{stft}(x_i, G(\hat{e}_i^c, \hat{e}_i^s, \hat{z}_i))]. \quad (14)$$

Knowledge Distillation Loss: At the same time, it is also expected that the representations of different classes in the reconstructed latent vectors $E_{MI}(\hat{x}_i^c)$, and $E_{MI}(\hat{x}_j^c)$ are apart, when $y_i \neq y_j$. However, calculating the cross-entropy loss with hard labels on generated samples would not be ideal since the generation quality of the generator is low in the early stages. Therefore, we adopt teacher-student knowledge distillation inspired by [26] to assign soft labels to the generated samples dynamically. We use baseline models (i.e., the same structure as $C_{MI} \circ E_{MI}$ and $C_{sbj} \circ E_{sbj}$) trained with the same real training set with hard labels as the teaching classification model T_{MI} and T_{sbj} . The encoders E_{MI}, E_{sbj} are then forced to minimize the KL divergence between the probability distribution predicted by classifiers C_{MI}, C_{sbj} and that predicted by the teaching models T_{MI}, T_{sbj} :

$$\begin{aligned} \mathcal{L}_{MI}^{KL} &= KL(C_{MI} \circ E_{MI}(\hat{x}_i^c) || T_{MI}(\hat{x}_i^c)), \\ \mathcal{L}_{sbj}^{KL} &= KL(C_{sbj} \circ E_{sbj}(\hat{x}_i^c) || T_{sbj}(\hat{x}_i^c)). \end{aligned} \quad (15)$$

E. Overall Training

Based on the training strategies, the overall training losses are $\mathcal{L}_{Gen}$ and $\mathcal{L}_{Dis}$. The generative loss $\mathcal{L}_{Gen}$ is a weighted sum of the losses $\{\mathcal{L}_{Recon}^S, \mathcal{L}_{KL}^S, \mathcal{L}_{Gen}^S, \mathcal{L}_{MI}^S, \mathcal{L}_{sbj}^S, \mathcal{L}_{Cycle}^S, \mathcal{L}_{Gen}^{CI}, \mathcal{L}_{Gen}^{CC}, \mathcal{L}_{CycleE}^{CC}, \mathcal{L}_{CycleX}^{CC}, \mathcal{L}_{MI}^{KL}, \mathcal{L}_{sbj}^{KL}\}$. The discriminative loss $\mathcal{L}_{Dis} = \mathcal{L}_{Dis}^S + \mathcal{L}_{Dis}^C$.

Over the course of training, both losses are minimized alternatively to update $\{E_{MI}, E_{sbj}, C_{MI}, C_{sbj}, G\}$ and D in an adversarial way. $\{w_k\}_{k=1}^{12}$ are the weights for loss functions.

In this work, we apply the WGAN-GP optimization strategy with loss change rate balancing from Ouyang et al. [49] to automatically set the update rate of the generator and discriminator instead of using a fixed ratio utilized in the original WGAN-GP

paper [45]. The adaptive ratio is determined by calculating the difference between the current and previous loss values. The differences are normalized by the loss magnitude [49]. The relative change losses r_G and r_D are defined by:

$$\begin{aligned} r_G &= |(\mathcal{L}_{Gen}^t - \mathcal{L}_{Gen}^{t-1}) / \mathcal{L}_{Gen}^{t-1}|, \\ r_D &= |(\mathcal{L}_{Dis}^t - \mathcal{L}_{Dis}^{t-1}) / \mathcal{L}_{Dis}^{t-1}|, \end{aligned} \quad (16)$$

where $\mathcal{L}^t$ and $\mathcal{L}^{t-1}$ represent the loss of current and previous iteration, and $|\cdot|$ represents the absolute value. A larger loss change ratio indicates that this component is in greater need of an update. During each iteration, if $r_D > \eta \cdot r_G$, the discriminator is updated and vice versa. Empirically, η was set up 5.

IV. EXPERIMENTS AND RESULTS

A. Experiment Setup

We implemented the whole framework with PyTorch. The MI information is highly entangled compared to subject-specific features. We used a ResNet structure with four residual convolution blocks, batch normalization, PReLU activation layers and dropout regularization with 0.3 drop probability for the encoder E_{MI} . In contrast, we used traditional convolution blocks without residual connections for E_{sbj} and E_z to save computational resources. For E_z , we removed the dropout layers and used pixel-wise normalizations instead of batch normalization layers [46]. The classifiers C_{MI} and C_{sbj} are made up of one fully connected layer. The generator G consists of two fully connected layers that process e^c and e^s vector. During the generation, The z vector is expanded to the original EEG channel number and concatenated with the outputs of the fully connected layers. A fusion network with skip connections and linear up-sampling layers was used to produce the final synthetic data. We used skip connections to ensure the effectiveness of the noise vector was not ignored in the deeper layers of the fusion network. No batch normalization layer or sigmoid activation was used as suggested by the original WGAN-GP paper [45]. Please refer to the details of our experiment settings and architectural designs in Section A of the Supplementary Materials.

B. Datasets

We evaluated our framework and baseline methods on three different MI datasets: BCIC IV 2a [50], PhysioNet [51] and OpenBMI dataset [52]. All subjects were healthy, and the ages were not reported in those specific datasets. The dataset details are as follows:

- **OpenBMI:** In this dataset, 62-channels signals were recorded from 54 different subjects referenced using the FCz channel [52]. Each subject was asked to imagine the grasping of the left hand or right hand, for 4 seconds in a total of 400 trials collected during two sessions on different days. We selected 20 channels located over the scalp region that corresponds to the motor cortex region (FC-5/3/1/2/4/6, C-5/3/1/z/2/4/6, and CP-5/3/1/z/2/4/6) in alignment with established practices in MI EEG studies [53]. The entire 4 s trial windows were resampled to 125 Hz resulting in a dimension of $54 \times 400 \times 20 \times 500$ (#subjects $\times$ #trials $\times$ #channels $\times$ #times points).
- **BCIC IV 2a:** This MI EEG dataset [50] consists of 22-channels EEG data recorded from 2 separate sessions of

9 different subjects using the left mastoid as the reference channel. The subjects were asked to imagine the movement of the left hand, right hand, both feet and tongue in 4-second trials. Similarly, we picked the data of left-hand and right-hand classes from 20 EEG channels that match the region of interest in OpenBMI (FC-3/1/z/2/4, C-5/3/1/z/2/4/6, CP-3/1/z/2/4, and P-1/z/2), to make our comparison more consistent across different datasets. The selected raw EEG data was then downsampled to 125 Hz resulting in a dimension of $9 \times 144 \times 20 \times 500$ (#subjects $\times$ #trials $\times$ #channels $\times$ #sampling-points).

- **PhysioNet:** It consists of 4-4.2 s trials of 64-channels EEG data recorded from 109 different subjects using both mastoids as references. The subjects were asked to imagine opening and closing either the left fist, right fist, both fists, or both feet in a single session of three runs [51]. We used the first 4 seconds in each trial and removed the data of subjects #88, #89, #92 and #100, because of the damaged recordings with multiple consecutive “rest” sections [11]. We picked the same 20 channels with the BCIC 2a dataset out of the 64 channels (FC-3/1/z/2/4, C-5/3/1/z/2/4/6, CP-3/1/z/2/4, and P-1/z/2), for consistent experimental design and comparison. The data was subsequently downsampled to 125 Hz resulting in a dimension of $105 \times 45 \times 20 \times 500$ (#subjects $\times$ #trials $\times$ #channels $\times$ #times points).

In practice, we applied the Braindecode (v0.5) library to load and preprocess the raw EEG data through bandpass filtering between 4-40 Hz using firwin filters with hamming windows, and downsampled to 128 Hz, followed by z-score normalization. We applied both **Leave-One-Subject-Out** for inter-subject and **Leave-One-Session-Out** for inter-session validation on the BCIC IV 2a and OpenBMI datasets. Whereas the PhysioNet MI EEG dataset was only used for **Leave-One-Subject-Out** for inter-subject validation, since it was only collected in a single session. In the Leave-One-Session-Out validation, we trained the model using all sessions from all participants, excluding the designated test session of the test participant. This approach was adopted due to our model’s larger size, requiring more data for effective learning.

C. Compared Methods

We compared our framework to the following methods based on their prominence and relevance in MI EEG, and the availability of their code.

- **Filter-Bank Common Spatial Pattern (FBCSP)** [6] applies band-pass filtering followed by spatial filtering using Common Spatial Pattern on each frequency band. SVM is used to perform classification based on the most discriminative CSP features from the filter bank.
- **DeepConvNet** [8] consists of a block with a temporal convolutional layer followed by a spatial convolutional layer and max pooling, and three blocks that each made up of a convolutional layer and a max-pooling layer.
- **ShallowConvNet** [8] consists of a temporal and a spatial convolutional layer comparable to the band-pass and spatial filtering in FBCSP, and followed by a squaring non-linearity, a mean pooling layer, and a logarithmic activation function.

TABLE I
AVERAGE INTER-SUBJECT CLASSIFICATION PERFORMANCE ON DIFFERENT DATASETS

Methods	BCIC 2a				PhysioNet				OpenBMI			
	Acc	F1	Precision	Recall	Acc	F1	Precision	Recall	Acc	F1	Precision	Recall
FBCSP [6]	59.0 ± 9.0	62.5 ± 14.5	59.4 ± 10.5	74.6 ± 25.2	60.0 ± 3.3	66.9 ± 2.5	56.9 ± 2.7	81.3 ± 5.2	60.2 ± 3.4	44.4 ± 9.8	74.9 ± 9.3	33.0 ± 10.3
DeepConvNet [8]	80.3 ± 6.8	78.9 ± 11.9	85.3 ± 4.7	77.0 ± 17.7	70.5 ± 3.1	60.8 ± 3.3	61.4 ± 3.3	61.8 ± 5.1	74.3 ± 2.4	74.0 ± 3.0	74.9 ± 6.4	75.6 ± 8.0
ShallowConvNet [8]	76.2 ± 12.4	75.9 ± 16.1	80.9 ± 16.0	80.0 ± 22.7	68.2 ± 4.2	57.8 ± 6.1	61.8 ± 5.5	58.6 ± 14.5	75.0 ± 4.0	74.4 ± 3.7	75.5 ± 6.2	75.4 ± 5.5
EEGNet [9]	73.8 ± 11.6	72.1 ± 17.8	77.4 ± 11.9	74.1 ± 23.1	72.6 ± 2.7	62.5 ± 2.9	64.2 ± 5.0	63.3 ± 7.9	72.6 ± 3.3	72.0 ± 3.1	73.9 ± 6.6	72.4 ± 5.6
ResNet [54]	71.7 ± 10.5	71.1 ± 11.9	74.9 ± 24.4	74.8 ± 14.4	67.8 ± 1.7	67.7 ± 3.7	69.7 ± 0.5	67.5 ± 3.3	72.4 ± 3.0	72.0 ± 4.5	72.0 ± 9.4	72.8 ± 2.6
CRAM [11]	70.3 ± 14.4	62.3 ± 18.2	68.0 ± 14.3	65.2 ± 15.3	64.7 ± 2.4	60.2 ± 4.0	61.0 ± 4.2	60.6 ± 4.1	69.8 ± 4.2	67.5 ± 4.6	67.9 ± 4.7	67.6 ± 4.6
GCRAM [12]	67.9 ± 13.0	61.2 ± 16.7	65.4 ± 13.3	63.9 ± 13.8	62.4 ± 3.0	57.6 ± 2.9	58.2 ± 3.2	57.9 ± 3.0	68.6 ± 4.3	65.0 ± 4.1	65.3 ± 3.9	65.2 ± 4.0
GCRAM* [12]	/	/	/	/	70.6 ± 3.4	67.7 ± 3.6	68.0 ± 3.4	67.8 ± 3.6	/	/	/	/
MIN2NET [39]	60.1 ± 3.5	54.7 ± 7.3	62.0 ± 4.5	50.4 ± 10.7	63.0 ± 2.1	61.8 ± 2.3	63.1 ± 2.3	60.6 ± 3.6	65.6 ± 4.1	64.3 ± 6.6	66.4 ± 4.6	63.4 ± 10.3
EEG Conformer [55]	70.9 ± 10.7	68.0 ± 18.1	70.6 ± 24.5	75.7 ± 15.6	69.4 ± 3.5	66.9 ± 7.0	64.0 ± 2.1	71.4 ± 3.3	70.5 ± 3.0	70.7 ± 3.8	71.9 ± 8.6	70.4 ± 4.9
w/o E_z	68.7 ± 7.7	63.1 ± 12.1	67.8 ± 20.8	69.0 ± 19.2	66.0 ± 4.9	64.7 ± 5.1	64.3 ± 2.8	68.0 ± 10.2	71.6 ± 2.5	70.1 ± 5.6	67.5 ± 3.9	74.3 ± 5.2
Ours	83.8 ± 6.4	83.0 ± 7.3	81.7 ± 10.5	85.1 ± 6.2	73.1 ± 3.6	74.5 ± 4.2	79.1 ± 7.8	70.6 ± 4.7	76.6 ± 3.5	76.7 ± 3.6	76.9 ± 4.3	76.5 ± 4.0

*The GCRAM version utilizing the whole 64 EEG channels of the PhysioNet dataset.

TABLE II
AVERAGE INTER-SESSION CLASSIFICATION PERFORMANCE ON DIFFERENT DATASETS

Methods	BCIC 2a				OpenBMI			
	Acc	F1	Precision	Recall	Acc	F1	Precision	Recall
FBCSP [6]	64.4 ± 11.6	66.8 ± 16.4	64.5 ± 16.3	77.5 ± 25.7	62.1 ± 12.5	61.1 ± 19.2	64.8 ± 14.2	68.6 ± 30.8
DeepConvNet [8]	85.0 ± 8.0	85.8 ± 9.7	90.7 ± 8.5	83.1 ± 12.5	63.7 ± 9.5	62.2 ± 16.5	60.8 ± 11.5	68.4 ± 24.2
ShallowConvNet [8]	85.2 ± 9.2	83.8 ± 13.6	82.7 ± 13.8	88.8 ± 10.5	67.6 ± 13.6	61.7 ± 24.2	65.5 ± 19.8	63.9 ± 30.7
EEGNet [9]	80.5 ± 10.9	80.3 ± 11.8	79.5 ± 14.0	79.7 ± 17.5	67.1 ± 14.1	57.5 ± 24.8	63.7 ± 21.4	57.8 ± 30.7
ResNet [54]	77.6 ± 12.6	76.0 ± 14.8	75.1 ± 20.1	81.5 ± 12.5	63.2 ± 8.2	60.4 ± 14.3	61.7 ± 24.4	65.2 ± 8.3
CRAM [11]	77.3 ± 13.4	73.1 ± 14.2	74.9 ± 12.4	74.0 ± 12.9	66.7 ± 8.6	59.4 ± 10.6	62.1 ± 8.6	61.2 ± 8.7
GCRAM [12]	73.6 ± 12.4	68.9 ± 14.0	70.8 ± 12.5	69.9 ± 12.8	64.8 ± 8.3	56.0 ± 12.1	58.1 ± 11.1	58.2 ± 9.7
MIN2NET [39]	66.2 ± 6.5	63.3 ± 9.6	67.4 ± 6.4	60.8 ± 12.9	61.5 ± 6.4	61.6 ± 6.9	61.5 ± 6.3	62.2 ± 8.9
EEG Conformer [55]	78.1 ± 13.7	75.7 ± 17.7	73.7 ± 21.6	83.9 ± 14.8	67.9 ± 11.0	67.6 ± 14.9	71.7 ± 23.3	70.9 ± 13.5
Ours	88.3 ± 9.9	85.0 ± 10.0	92.5 ± 16.3	84.1 ± 12.4	77.4 ± 9.7	75.9 ± 11.8	74.1 ± 16.6	80.2 ± 9.9

- *EEGNet* [9] consists of a block with a 2D convolution filter and a depthwise convolution spatial filter, and a second block with a separable convolution operation.
- *ResNet* [54] consists of four residual convolution blocks, batch normalization, PReLU activation layers and dropout regularization. It has a similar structure as our E_{MI} encoder combined with C_{MI} classifier.
- *Convolutional Recurrent Attention Model (CRAM)* [11] first segments the input signal into temporal slices of equal lengths and processes the slices through a convolutional spatial-temporal encoding block, an LSTM block and a self-attention block. The encoding block consists of convolution followed by pooling operations and ELU activation functions.
- *Graph-based Convolutional Recurrent Attention Model (GCRAM)* [12] is an extension of the CRAM model. The input EEG signal for GCRAM is first multiplied with an electrode adjacency matrix before it is segmented into temporal slices. We used the neighboring connection adjacency matrix during testing. The adjacency matrix captures the spatial relationship between sensors, where the geometric neighbors are connected.
- *MIN2NET* [39] integrates deep metric learning into a multi-task CNN-autoencoder to derive a compact and discriminative latent representation from EEG data. It minimizes an MSE loss, a triplet loss, and a cross-entropy loss during training simultaneously.
- *EEG Conformer* [55] combines CNN with Transformer to learn the spatial and temporal representations and finally perform the classification with two fully connected layers.

Its CNN block shares a similar structure with the ShallowConvNet and the multi-head attention was applied along the window (temporal) dimension.

D. Quantitative Results

1) *Inter-Subject/Session Validation*: The inter-subject classification and inter-session results are presented in Tables I and II, respectively. As shown, our framework achieved the highest average classification performance in all datasets and both experimental protocols, with a low standard deviation. This indicates that our framework is more robust and consistent in the challenging cross-subject/session tasks.

It should be noted that although some baseline models (e.g. EEGNet) outperformed other more recent sophisticated algorithms (e.g. CRAM, GCRAM) in our binary classification experiment, they might not perform as well in multi-class tasks, as reported in [39]. This might suggest that faced with the intrinsic complexity of cross-subject/session experiments, simple CNNs like EEGNet [9] are effective for MI EEG data, while advanced methods like GCRAM might need dataset-specific tuning. Particularly, the limited number of channels might affect the performance of GCRAM, since the original paper [12] leveraged the whole channels to construct the adjacency matrix as well as fuse features. We also reported the results of GCRAM with the full 64 channels on the PhysioNet dataset while keeping all other training parameters and pre-processing methods the same, denoted as GCRAM*. However, it still shows relatively limited generalization performance.

2) *Quantitative Validation on the Realism of Synthetic Data*: To quantitatively validate the effectiveness of the data generation part, we applied two schemes to evaluate the realism of the

TABLE III
BASELINE CLASSIFICATION ACCURACY WITH AND WITHOUT DATA AUGMENTATION BY THE SYNTHETIC DATA

Models	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Fold 6	Fold 7	Fold 8	Fold 9	Average
DeepConvNet	69.2	76.9	78.3	76.0	71.3	76.4	75.1	74.3	71.3	74.3
EEGNet	65.7	77.2	77.5	74.6	70.2	75.8	73.8	70.7	68.0	72.6
DeepConvNet + aug	71.8	81.4	76.3	76.3	72.9	80.1	79.8	74.8	74.2	76.4
EEGNet + aug	71.5	78.3	72.8	77.3	69.6	79.1	69.3	72.3	69.8	73.3

TABLE IV
ACCURACY AND KL DIVERGENCE OF THE DIFFERENT SETTINGS FOR SYNTHETIC DATA QUALITY VALIDATION.

Train-Test Split	Accuracy $\uparrow$		KL Divergence $\downarrow$	
	MI	subject	MI	subject
tr-real-te-fake	97.2 $\pm$ 1.5	92.5 $\pm$ 2.2	0.1354 $\pm$ 0.003	0.1636 $\pm$ 0.002
tr-fake-te-real	90.5 $\pm$ 2.6	85.2 $\pm$ 1.7	0.1523 $\pm$ 0.004	0.1742 $\pm$ 0.003
tr-real-te-real	98.0 $\pm$ 1.0	98.0 $\pm$ 1.5	0.1378 $\pm$ 0.002	0.1169 $\pm$ 0.002

Accuracy: higher is better, KL divergence: lower is better.

generated synthetic samples. One is to train a baseline classifier for the classification task and to check whether the performance can be lifted supplemented with additional generated samples. The other is to perform cross-validation between real and synthetic samples, and to observe the model performance change once the data domain is shifted.

a) Baseline Model Augmentation with Synthetic Data: We conducted an experiment to evaluate the potential improvement of baseline methods when trained with the augmentation enabled by the generated synthetic data. Table III shows the inter-subject classification results of two baseline classification models DeepConvNet and EEGNet with and without augmentation. In the augmentation case, an extra synthetic copy of the training data generated by our model was used for training. As the results shown in Table III, both baseline classifiers benefited from the augmentation with the synthetic samples generated by our proposed method.

b) Test-Real-Train-Fake and Train-Fake-Test-Real: In addition, we trained a DeepConvNet classifier by training with the real data and testing on fake data, denoted as the tr-real-te-fake test. The reversed setting is denoted as tr-fake-te-real. We compared the reported results with the tr-real-te-real setting, where the training and the testing subsets were picked from different trials. We calculated the average accuracy and KL divergence between the real train and validation sets for the tr-real-te-real test case as a reference. As shown in Table IV, the baseline model can achieve over 90% accuracy in both cases for motor imagery classification and over 85% accuracy in both cases for subject classification. We also measured the KL divergence between the probability distributions of two predictions, The KL divergences for MI classifications are comparable to the tr-real-te-real, while the KL divergences for subject prediction are slightly higher. The results also indicate that the fake data is more similar to the real data in terms of motor imagery information than subject information.

E. Qualitative Results

1) Time-Domain Visualization: To inspect the fake data visually, we presented in Fig. 6 an example consisting of a real sample and one fake sample generated from the e^c and e^s of that real one. Both the time-domain representations and brain activation topographical maps across frequency bands are presented.

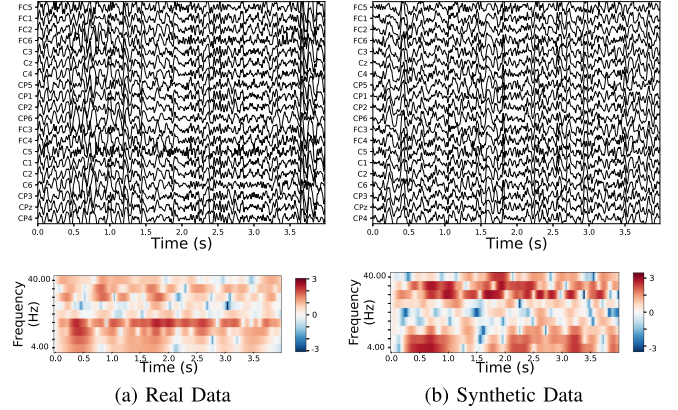


Fig. 6. Samples of real and synthetic data. Left: Real data, Right: Synthetic data; Top: Time-domain representations, Bottom: time-frequency plot of Cz channel.

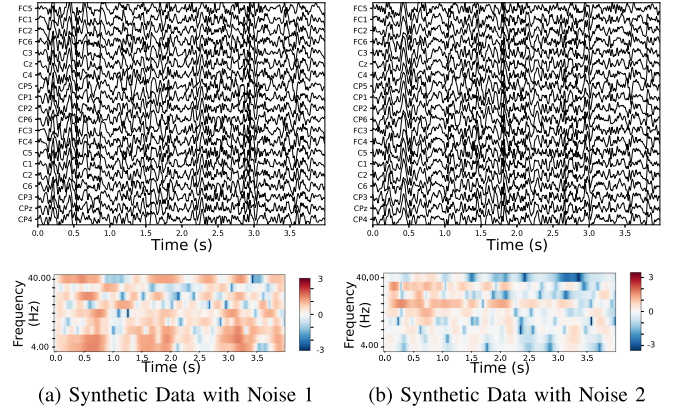


Fig. 7. Synthetic data with different noise vectors n sampled from $\mathcal{N}(0, 1)$. Left: Synthetic data with noise 1, Right: synthetic data with noise 2; Top: Time-domain representations, Bottom: time-frequency plot of Cz channel.

The generated fake data contains similar temporal segments with different temporal orders from the real data, which indicates the generated data is both realistic and diverse. Furthermore, we present the fake data generated with different noise vectors in Fig. 7. The two signals (a) and (b) are different in amplitude and temporal order, which indicates that the noise vector can induce variances in the generated data.

2) Frequency-Domain Visualization: We also analyze the generated data quality in the frequency domain. The average power spectrum density of all data and channels is shown in Fig. 8(a). The mean and standard deviation are similar between the real and fake data. However, as shown in Fig. 8(b), the frequency distribution of the real data in blue is different from

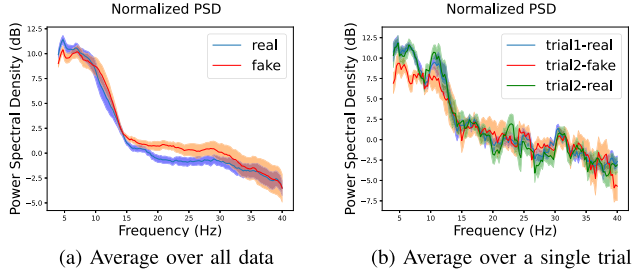


Fig. 8. Average power spectrum density of all channels.

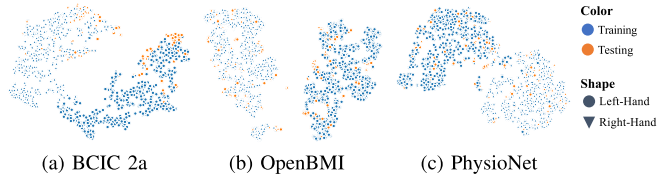


Fig. 9. t-SNE visualizations of latent e^c extracted from the MI encoder E_{MI} over different datasets. Blue: Training subjects, Orange: Test subjects, Circle: Left-hand, Triangle: Right-hand. Please enlarge or refer to https://xiaogu.site/eeg_disentangle for better quality.

the fake data in red for single trials. The real signal of a different trial of the same subject is shown in green color. The variations between real and fake data are similar to how the real trial 2 (green) would vary compared to real trial 1 (blue). Therefore, the generated data is both diverse and realistic in the frequency domain.

3) *t-SNE Visualizations of Final Feature Layers of Different Datasets*: Fig. 9 shows the t-SNE projection of the learned MI embedding features of all the datasets. The features were extracted from an inter-subject classification task. It can be observed that most of the features from the test subject data are clustered correctly in the two MI classes instead of being in the middle. It indicates that our MI encoder successfully learned the MI features from all three datasets without subject-specific information.

F. Additional Results and Further Discussions

1) *Ablation Study*: Additionally, we conducted experiments with the OpenBMI dataset to determine the importance of losses and objectives. We conducted paired student t-tests with Bonferroni correction to evaluate the significance of the results. For the setting without the classifiers, we trained a separate classification layer with the encoded features from the MI encoder after the network was trained. As shown in Fig. 10(a), the network achieved significantly lower accuracy without the classifiers. This suggests that the subject classifier and the task classifier are critical to letting the encoders separately encode specific information respectively. Furthermore, the network failed to converge without updating the GAN adversarial losses. The results suggest that the GAN structure is an important component of our framework. With only the STFT losses, the generator is prone to mode collapses.

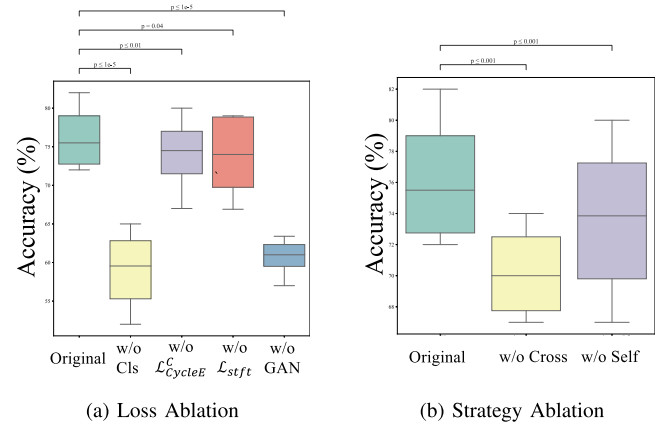


Fig. 10. Results of the ablation of different losses and training strategies on the OpenBMI dataset.

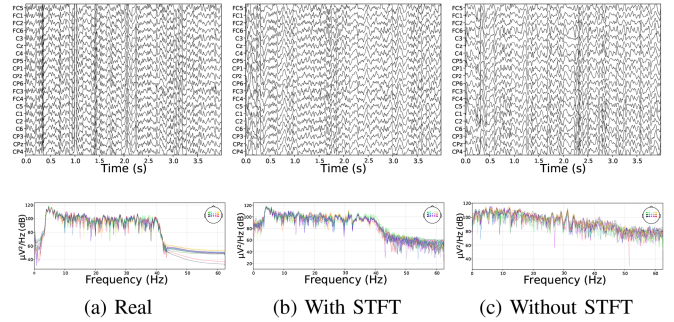


Fig. 11. Real and synthetic data trained with/without STFT losses in both time and frequency domains.

The results in Fig. 10(b) show the ablation of our training strategies. Our framework achieved significantly higher accuracy with both cross-subject generation objectives and self-generation objectives. The results suggest that the objectives are essential to disentangle the representations completely.

2) *Importance of Noise Factorization*: Compared to the conventional disentangled representation learning framework, which normally consists of two encoders, our architectures are highlighted by factorizing the noise out by a third encoder. To validate the effectiveness of such a design, we conducted a series of experiments with conventional disentangled representation learning. A model without the noise encoder branch was used as a comparison. The generator generates samples using the latent vectors of MI and subject only. The losses $\{\mathcal{L}_{KL}^S, \mathcal{L}_{Gen}^S\}$ were not used for training. Its results are presented in Table I, marked as $w/o E_z$. The inferiority of this method highlights the importance of considering the noise branch in EEG representations.

3) *Importance of Noise Losses*: As discussed in Section II-I-C, we applied a KL divergence loss and a latent consistency loss to ensure the functionality of the latent vector z . In other words, it can generate synthetic data with variances by sampling different n . In Fig. 12, we show two t-SNE plots of the noise encoder comparison with/without using $\mathcal{L}_{KL}^S$ and $\mathcal{L}_{Cycle}^S$. It is shown that minimizing these losses during the self-generation/reconstruction task allows the noise encoder to learn a randomly distributed noise independent of the given

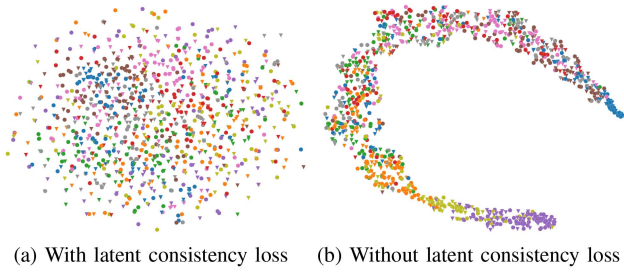


Fig. 12. t-SNE visualizations of final feature layers of noise encoders with/without latent consistency loss. Different colors represent different subjects. Regarding Shapes, Circle: Left-hand, Triangle: Right-hand. Please enlarge or refer to https://xiaogu.site/eeg_disentangle for better quality.

TABLE V
LINEAR CLASSIFICATION BASED ON EXTRACTED FEATURES

e^c	e^s	e^z	MI (%)	Subj (%)
✓			89.5	69.0
	✓		59.0	99.5
		✓	52.5	57.5
✓	✓		87.5	99.0
	✓	✓	59.5	99.5
✓		✓	89.0	78.5
✓	✓	✓	88.0	99.0

subject or MI. In contrast, without minimizing them, E_z encodes some subject-specific information since the features of the same subject (same color) get closer together as shown in Fig. 12. The noise encoder that learns subject-specific information reduces thus affecting the disentanglement.

4) Importance of STFT Losses: As discussed in Section II-I-C1, we used the STFT loss to ensure our model can generate data that is similar to the original data in frequency domains. We evaluated the importance of the STFT losses by training another model without the STFT losses while keeping all other setups the same. Model collapse was observed after training as the generation qualities are only controlled by the discriminator. In addition, a sample before the collapse is shown in Fig. 11. Although the synthetic data reconstructed by the model trained without STFT losses seems acceptable in the time domain, the data failed to capture key characteristics in the frequency domain, such as the effect of the bandpass filter. Furthermore, the model trained without STFT losses failed to generate realistic and diverse data from different subjects, motor imagery classes, or noise vectors. Thus, loss functions in the frequency domain, such as STFT loss, are essential for GAN training stability for EEG generation.

5) Extracted Features: We conducted an experiment to analyze the extracted features, e^c , e^s and e^z from the three separate encoders. Simple linear classifiers were trained to predict the MI or subject labels using different combinations of concatenated features. As shown in Table V, The linear classifier trained with only the MI feature e^c achieved the highest classification accuracy for MI prediction, but lower accuracy for subject prediction. On the other hand, the linear classifier trained with subject feature e^s achieved low accuracy for MI prediction but high accuracy for subject prediction. Furthermore, the linear classifier trained using noise feature e^z achieved nearly random accuracy for both MI and subject prediction. The experiment results

further confirmed the fact that our three encoders successfully disentangled the MI, subject and noise features in EEG data.

In addition, considering the constraints of real-world BCI applications, where devices typically have fewer channels and limited computational resources, we have also conducted additional experiments to assess the computational cost and the adaptability of our approach under limited channel configurations and different BCI task types. Detailed results of these evaluations can be found in the Supplementary Materials.

V. CONCLUSION

The generalization capability of motor imagery classification models has been a long-lasting challenge for real-world BCI applications. They mostly require long-term tedious calibration procedures to adapt to the new sessions or subjects, which restricts the practicality for real-world deployment. Furthermore, conventional deep learning methods require a large amount of data to train, and the generic training pipelines suffer from variations between different subjects and sessions. In this work, we developed a noise-factorized disentangled representation learning framework that aims to enhance the performance of handling cross-session/subject variations. We proposed a novel joint discriminative and generative framework for MI EEG representation disentanglement, which integrates conditional variational autoencoder and generative adversarial training. It functions by leveraging the fundamental loss functions for discriminative and generative training, as well as our proposed novel training strategies. The results show that our proposed framework can generate both realistic and diverse EEG signals of specific classes, and the generated data can be used to improve the performance of baseline classifiers. More importantly, it is able to extract subject/session-independent and task-specific features, thus facilitating cross-subject/session generalization.

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